

Enhancing AI Education Through Puzzle-Based Learning

Integrating IBM SkillsBuild Chatbot Course with IBM Granite

2nd Group Meeting
MSc Advanced Computing
King's College London
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Problem & Research Questions

Problem

- Online learning has low engagement — easy to pause, hard to stay focused
- Games like "while True: learn()" teach AI, but lack adaptive support
- LLM tutors can answer questions, but lack structure
- **Gap: Limited research merging game-based learning with LLM tutoring**

Research Questions

- RQ1: Can a game-based approach effectively teach chatbot/NLP concepts?
- RQ2: Does LLM tutoring support improve the learning experience?

Aim: Develop a narrative puzzle game teaching IBM SkillsBuild chatbot course, with IBM Granite LLM as an in-game tutor.

Game Concept: Help a Friendly Ghost Communicate

Story Premise

- Player moves into a new apartment and discovers a friendly ghost
- The ghost is trapped in the smart home system and wants to communicate
- But it doesn't understand human language — player must help it learn
- **Helping the ghost communicate = Learning to build a chatbot**

Characters

- Ghost: Cute, friendly, wants to communicate but can't — no name yet
- Lily: Ghost expert, guides the player (powered by IBM Granite LLM)
→ Professional but friendly, has helped many ghosts before
- Player: Just moved in, learns chatbot concepts by helping the ghost

Story Structure → Course Content Mapping

Act	Story	Learning Content	Puzzle Type
Act 1	Discover the ghost Lily appears to help	What is a chatbot? Why is communication hard?	Identify problem
Act 2	Try to understand the ghost's messages	NLP basics Intent classification	Classify ghost's message intents
Act 3	Build a communication bridge with the ghost	Chatbot architecture Components	Connect components (node-based)
Act 4	Teach the ghost how to respond	Building chatbot Testing & debugging	Train and fix errors
Act 5	Ghost can communicate! Player names it	Review all concepts Warm ending	Final conversation

Tools & Technologies

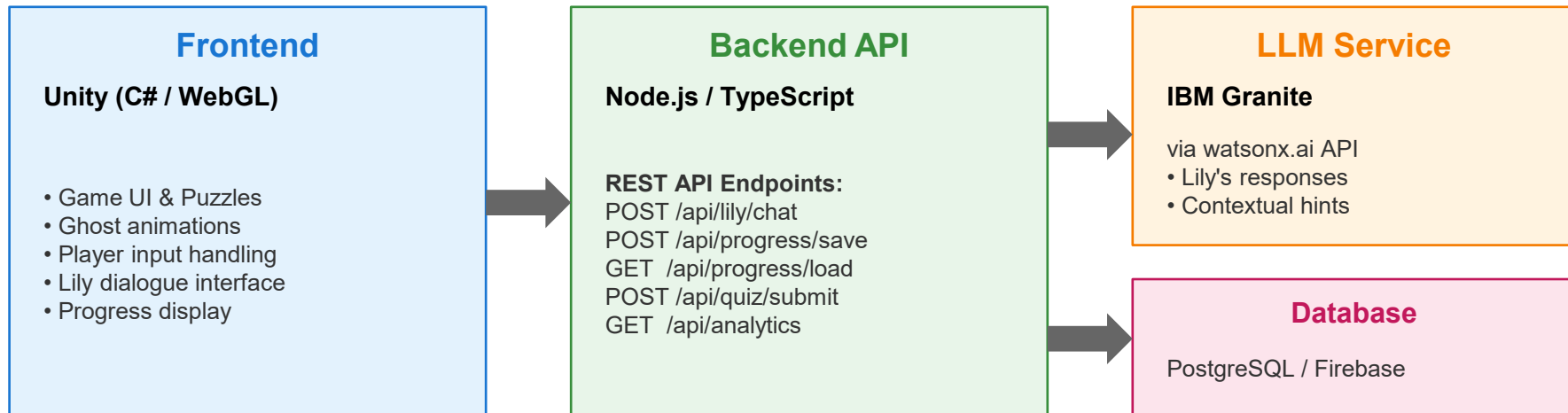
Category	Tool	Example Usage in This Project
Game Engine	Unity (C#)	Ghost animations, puzzle UI, player input handling, WebGL deployment
Backend	TypeScript / Node.js	API endpoints for Lily dialogue, player progress sync, IBM Granite integration
LLM	IBM Granite	Lily provides hints when stuck, answers NLP/chatbot questions, adapts to player level
Database	PostgreSQL / Firebase	Store completed acts, quiz scores, time spent per puzzle
Version Control	Git / GitHub	Track changes, collaborate, backup code
Design	Figma	Ghost character design, puzzle wireframes, Lily dialogue box UI

All tools are free or available through King's / IBM partnership.

System Architecture

Unity (Frontend) → Node.js API → IBM Granite (LLM) + Database (Progress)

System Architecture



Data Flow Example: Player asks Lily for help

1. Player clicks "Ask Lily" button in Unity
2. Unity sends POST /api/lily/chat with { message, currentAct, puzzleState }
3. Backend constructs prompt with course context → calls IBM Granite API
4. Granite returns response → Backend logs interaction to Database
5. Backend returns Lily's response → Unity displays in dialogue box

Literature Review Summary

	Paper 1 (Deng et al.)	Paper 2 (Chen & Chang)
Type	Systematic Review + Meta-analysis	Experimental Study (RCT)
Focus	ChatGPT in education (general)	AI + Game-based learning
Sample	69 studies reviewed	202 students, 3 groups
Key Finding	ChatGPT improves learning, motivation, reduces cognitive load	Game+AI > Game alone, AI+examples works best
Journal IF	~12 (Computers & Education)	~5.5 (Education & Info Tech)

Research Gap My Project Addresses

- Few studies combine narrative game mechanics + LLM tutoring for AI/CS education
- No existing game teaches IBM SkillsBuild chatbot course content
- **My project: Story-driven puzzle game + IBM Granite tutor → fills both gaps**

Evaluation Methodology

Learning Analytics (Automatic)

Data collected automatically during gameplay:

- Time spent per Act / Puzzle
- Number of attempts per puzzle
- Frequency of asking Lily for help
- Quiz scores and error patterns
- Completion rate

How This Answers Research Questions

Research Question	Data Source
RQ1: Can game teach chatbot concepts?	Pre/Post test score comparison
RQ2: Does Lily (LLM) improve learning?	Help frequency vs completion rate, interview feedback

Current Progress & Challenges

Completed

- Project scope defined
- IBM SkillsBuild course selected
- Technology stack decided
- Game concept and story designed
- Zotero library set up

In Progress

- Learning Unity basics
- Detailed course content analysis
- Literature review expansion
- Preliminary report writing

Challenges / Questions for Discussion

- IBM watsonx.ai access: What is the process to apply?

Future Work

Immediate Next Steps

- Apply for IBM watsonx.ai access
- Complete Unity fundamentals tutorial
- Create a detailed game design document with a story script

Thank You